

➤ **AI in Finance: Threats and Challenges**

Seminar SS 2025

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Agenda

- Motivation
- Topics of the seminar
- Seminar organization

➤ Motivation

- AI in finance is applying Artificial Intelligence technologies to improve financial services and processes.

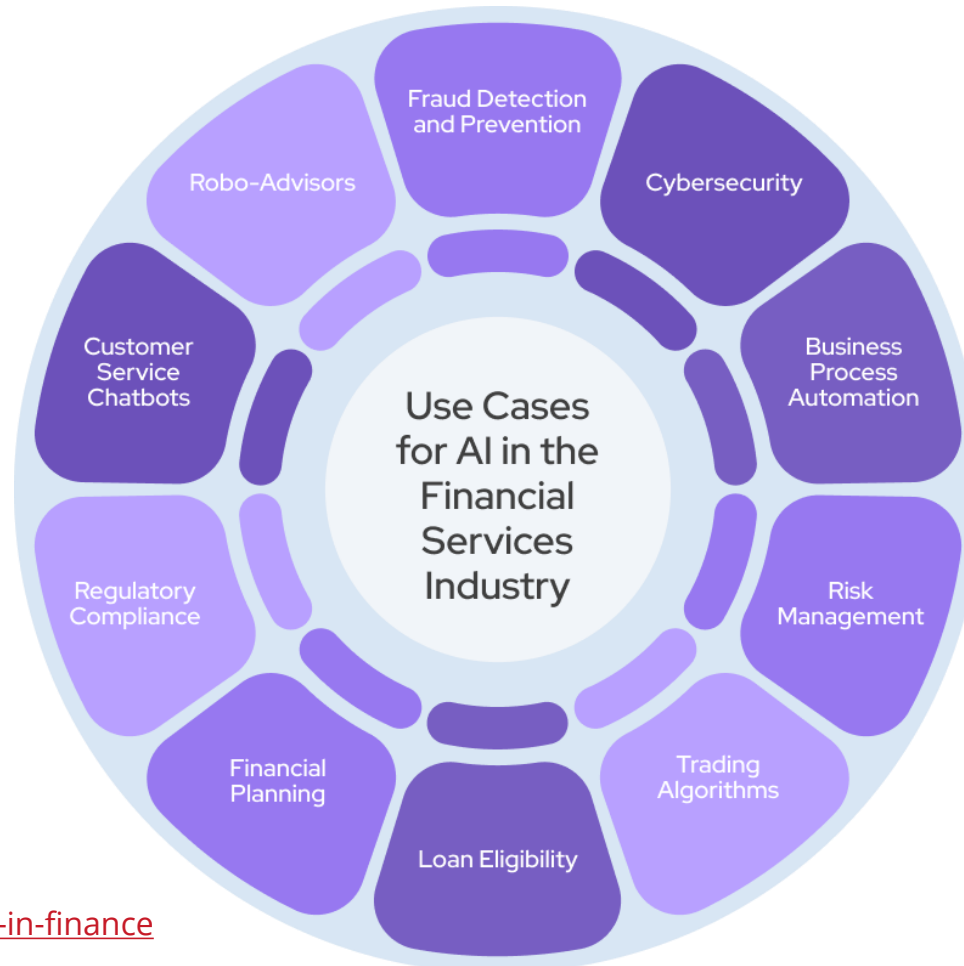


Image: <https://www.aifalabs.com/blog/ai-in-finance>

- Benefits
 - Improving decision-making process
 - Algorithmic trading improvement
 - Improving financial forecasting
 - Improving compliance and fraud detection
 - Reducing illegal insider trading
 - Reducing operational costs
 - Customer experience enhancement
 - ...

- Threats and challenges
 - Data privacy challenges
 - Data quality and data availability
 - Fairness of AI-based models
 - Explainability and transparency of AI-based models
 - Selecting the optimal machine learning model
 - Accountability of AI output
 - Job displacement
 - ...

Objective of the seminar

- Explore and exploit the advanced applications and the challenges of AI in finance
 - Recognize the threats and challenges
 - Explore the practical examples and case studies
 - Investigate the strategies to address those challenges

➤ **Topics of the seminar**

Suggested topics

Select a suggested topic or propose your topic related to the theme of the seminar

1. Data privacy in AI-powered finance
2. Data quality challenge in finance
3. Big Data and finance
4. Dimensionality reduction for financial data
5. Clustering for financial data analysis
6. Synthetic data in finance
7. Generative AI in finance
8. Neural networks in finance
9. Explainability and transparency of AI models
10. Bias and fairness of AI models
11. Reinforcement learning in finance
12. Large language models in finance
13. Algorithmic trading
14. AI and fraud detection
15. AI and credit scoring

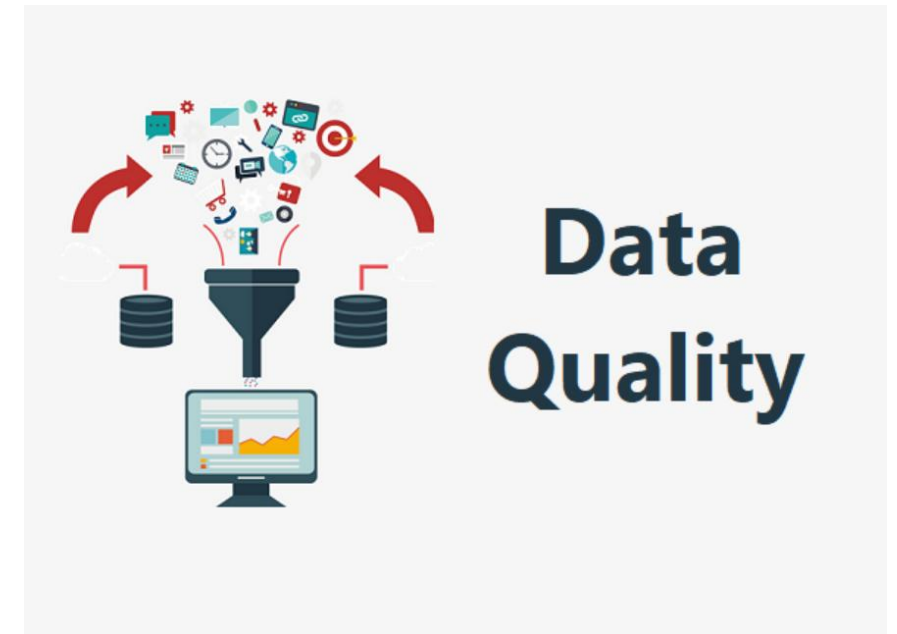
1. Data privacy in AI-powered finance

- Explore the challenges and solutions related to data privacy in AI-powered finance, focusing on regulatory compliance, and technological safeguards.
- Evaluate how financial institutions can balance innovation with user privacy protection while mitigating risks associated with AI-driven data processing.
- Related work:
 - Tripathi, M., & Mukhopadhyay, A. (2020). Financial loss due to a data privacy breach: An empirical analysis. *Journal of Organizational Computing and Electronic Commerce*, 30(4), 381-400.
<https://doi.org/10.1080/10919392.2020.1818521>
 - Gai, K., Qiu, M., Sun, X., & Zhao, H. (2016, December). Security and privacy issues: A survey on FinTech. In *International Conference on Smart Computing and Communication* (pp. 236-247). Cham: Springer International Publishing. https://doi.org/10.1007/978-3-319-52015-5_24



2. Data quality challenge in finance

- Investigate the challenges of data quality in finance, focusing on issues such as accuracy, consistency, completeness, and timeliness.
- Identify the impact of poor data quality on financial decision-making and explore strategies to enhance data reliability and integrity.
- Related work:
 - Pipino, L. L., Lee, Y. W., & Wang, R. Y. (2002). Data quality assessment. *Communications of the ACM*, 45(4), 211-218.
<https://doi.org/10.1145/505248.506010>
 - Du, J., & Zhou, L. (2012). Improving financial data quality using ontologies. *Decision Support Systems*, 54(1), 76-86.
<https://doi.org/10.1016/j.dss.2012.04.016>



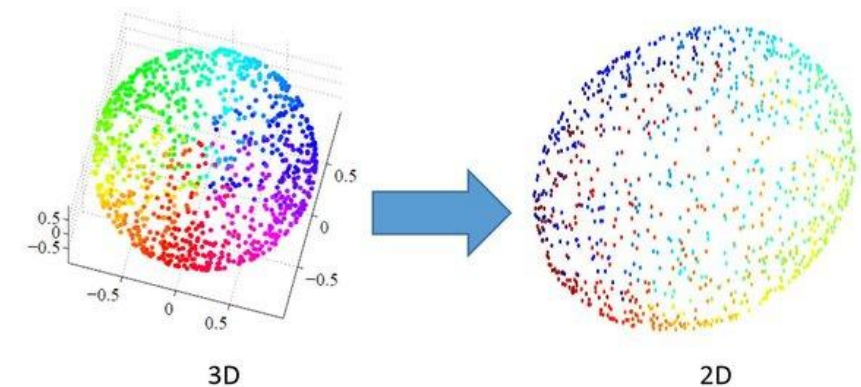
3. Big Data and finance

- Examine the challenges of Big Data in finance, including data management, security, and analytical complexity, etc.
- Explore how financial institutions can leverage Big Data effectively while addressing issues related to scalability, regulatory compliance, and decision-making accuracy.
- Related work:
 - Cockcroft, S., & Russell, M. (2018). Big data opportunities for accounting and finance practice and research. *Australian Accounting Review*, 28(3), 323-333.
<https://doi.org/10.1111/auar.12218>
 - Goldstein, I., Spatt, C. S., & Ye, M. (2021). Big data in finance. *The Review of Financial Studies*, 34(7), 3213-3225.
<https://doi.org/10.1093/rfs/hhab038>



4. Dimensionality reduction for financial data

- Explore dimensionality reduction techniques for financial data, focusing on improving data efficiency, reducing noise, and enhancing predictive accuracy.
- Evaluate multiple methods, such as PCA and autoencoders, to optimize data processing and support more effective financial analysis.
- Related work:
 - Sarlin, P. (2015). Data and dimension reduction for visual financial performance analysis. *Information Visualization*, 14(2), 148-167. <https://doi.org/10.1177/1473871613504102>
 - Bulut, C., & Arslan, E. (2024). Comparison of the impact of dimensionality reduction and data splitting on classification performance in credit risk assessment. *Artificial Intelligence Review*, 57(9), 252. <https://doi.org/10.1007/s10462-024-10904-1>



5. Clustering for financial data analysis

- Evaluate various clustering algorithms to enhance data-driven decision-making in finance.
- Investigate the challenges of applying clustering techniques to financial data analysis, including issues related to high dimensionality, data volatility, and interpretability.
- Explore potential solutions to improve clustering accuracy and effectiveness in financial decision-making.
- Related work:
 - Kou, G., Peng, Y., & Wang, G. (2014). Evaluation of clustering algorithms for financial risk analysis using MCDM methods. *Information sciences*, 275, 1-12.
<https://doi.org/10.1016/j.ins.2014.02.137>
 - Pattarin, F., Paterlini, S., & Minerva, T. (2004). Clustering financial time series: an application to mutual funds style analysis. *Computational Statistics & Data Analysis*, 47(2), 353-372.
<https://doi.org/10.1016/j.csda.2003.11.009>

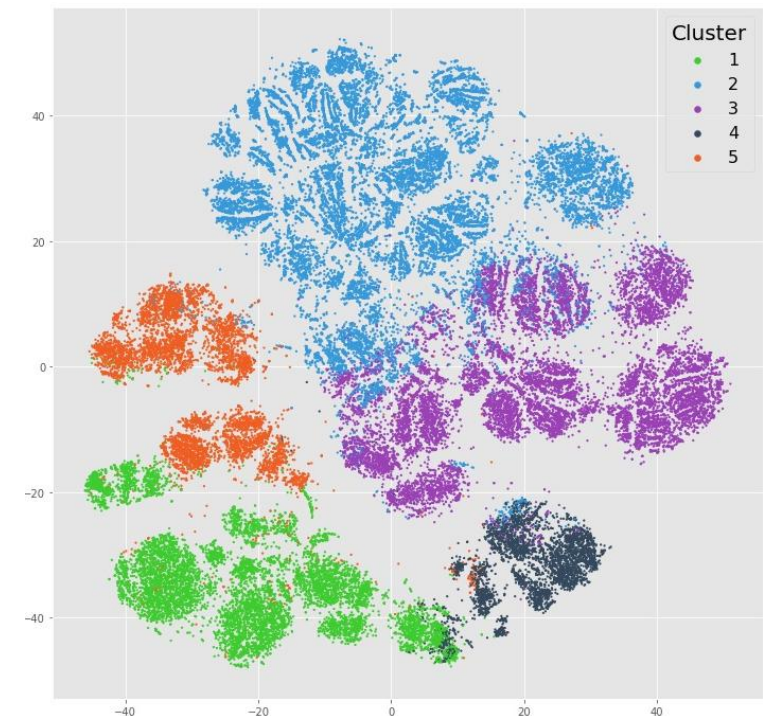
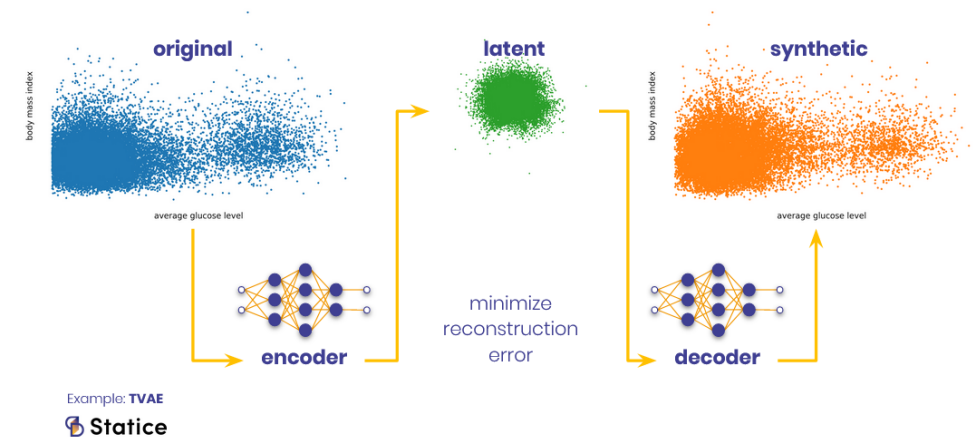


Image: <https://www.mdpi.com/1911-8074/14/2/50#>

6. Synthetic data in finance

- Explore the techniques to generate synthetic data in finance, focusing on its potential benefits for data privacy, model training, and risk management.
- Investigate the key challenges, including data fidelity, regulatory compliance, and reliability in real-world financial applications.
- Related work:
 - Assefa, S. A., Dervovic, D., Mahfouz, M., Tillman, R. E., Reddy, P., & Veloso, M. (2020, October). Generating synthetic data in finance: opportunities, challenges and pitfalls. *In Proceedings of the First ACM International Conference on AI in Finance* (pp. 1-8). <https://doi.org/10.1145/3383455.3422554>



7. Generative AI in finance

- Explore the applications of generative AI in finance, focusing on its potential for risk modeling, fraud detection, and automated decision-making.
- Investigate the key challenges, including data security, regulatory compliance, and model transparency in financial systems.
- Related work:
 - Chen, B., Wu, Z., & Zhao, R. (2023). From fiction to fact: the growing role of generative AI in business and finance. *Journal of Chinese Economic and Business Studies*, 21(4), 471-496.
<https://doi.org/10.1080/14765284.2023.2245279>
 - Desai, A. P., Ravi, T., Luqman, M., Mallya, G., Kota, N., & Yadav, P. (2024, December). Opportunities and challenges of generative-AI in finance. In *2024 IEEE International Conference on Big Data (BigData)* (pp. 4913-4920). IEEE.
<https://doi.org/10.1109/BigData62323.2024.10825658>



Image: <https://techbullion.com/generative-ai-in-finance-enhancing-financial-reporting-compliance-and-operational-efficiency-with-ai-agents/>

8. Neural networks in finance

- Explore the applications of neural networks in finance, focusing on their role in predictive modeling, algorithmic trading, and risk assessment.
- Examine the key challenges, including data quality, interpretability, computational complexity, and regulatory compliance.
- Related work:
 - McNelis, P. D. (2005). Neural networks in finance: gaining predictive edge in the market. *Academic Press*.
 - Li, Y., & Ma, W. (2010, October). Applications of artificial neural networks in financial economics: a survey. In *2010 International symposium on computational intelligence and design* (Vol. 1, pp. 211-214). IEEE.
<https://doi.org/10.1109/ISCID.2010.70>



9. Explainability and transparency of AI models

- Investigate the importance of explainability and transparency in AI models within the finance sector, focusing on their role in building trust and ensuring regulatory compliance.
- Address the challenges of making complex AI models interpretable while maintaining their predictive power and accuracy in financial applications.
- Related work:
 - Černevičienė, J., & Kabašinskas, A. (2024). Explainable artificial intelligence (XAI) in finance: a systematic literature review. *Artificial Intelligence Review*, 57(8), 216.
<https://doi.org/10.1007/s10462-024-10854-8>



10. Bias and fairness of AI models

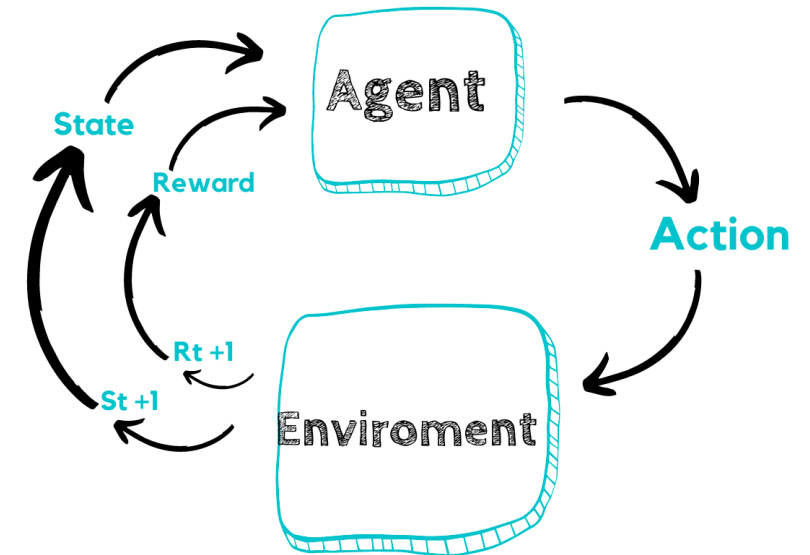
- Examine the issues of bias and fairness in AI models used in finance, focusing on how these biases can impact decision-making and financial outcomes.
- Explore methods for mitigating bias and ensuring that AI systems are fair, equitable, and compliant with ethical and regulatory standards in the financial sector.
- Related work:
 - McCalman, L., Steinberg, D., Abuhamad, G., Brunet, M. E., Williamson, R. C., & Zemel, R. (2022). Assessing AI fairness in finance. *Computer*, 55(1), 94-97.
<https://doi.org/10.1109/MC.2021.3123796>
 - Castelnovo, A., Crupi, R., Del Gamba, G., Greco, G., Naseer, A., Regoli, D., & Gonzalez, B. S. M. (2020, December). Befair: Addressing fairness in the banking sector. In *2020 IEEE international conference on big data (Big Data)* (pp. 3652-3661). <https://doi.org/10.1109/BigData50022.2020.9377894>



Image: <https://www.wavgroup.com/2023/12/22/real-ai-ai-fairness-is-a-huge-hurdle-ai-cant-count-fast-facts-top-headlines-and-quote-of-the-week/>

11. Reinforcement learning in finance

- Explore the applications of reinforcement learning in finance, particularly in areas such as portfolio optimization, trading strategies, and risk management.
- Investigate the key challenges, including data scarcity, model stability, and the need for real-time decision-making in complex financial environments.
- Related work:
 - Hambly, B., Xu, R., & Yang, H. (2023). Recent advances in reinforcement learning in finance. *Mathematical Finance*, 33(3), 437-503. <https://doi.org/10.1111/mafi.12382>
 - Charpentier, A., Elie, R., & Remlinger, C. (2021). Reinforcement learning in economics and finance. *Computational Economics*, 1-38. <https://doi.org/10.1007/s10614-021-10119-4>



12. Large language models in finance

- Explore the applications of large language models in finance, focusing on their potential for automating tasks such as customer support, sentiment analysis, and financial forecasting.
- Investigate the challenges of deploying these models in finance, including data privacy, model interpretability, and regulatory compliance.
- Related work:
 - Li, Y., Wang, S., Ding, H., & Chen, H. (2023, November). Large language models in finance: A survey. In *Proceedings of the fourth ACM international conference on AI in finance* (pp. 374-382).
<https://doi.org/10.1145/3604237.3626869>
 - Lee, J., Stevens, N., & Han, S. C. (2025). Large Language Models in Finance (FinLLMs). *Neural Computing and Applications*, 1-15.
<https://doi.org/10.1007/s00521-024-10495-6>



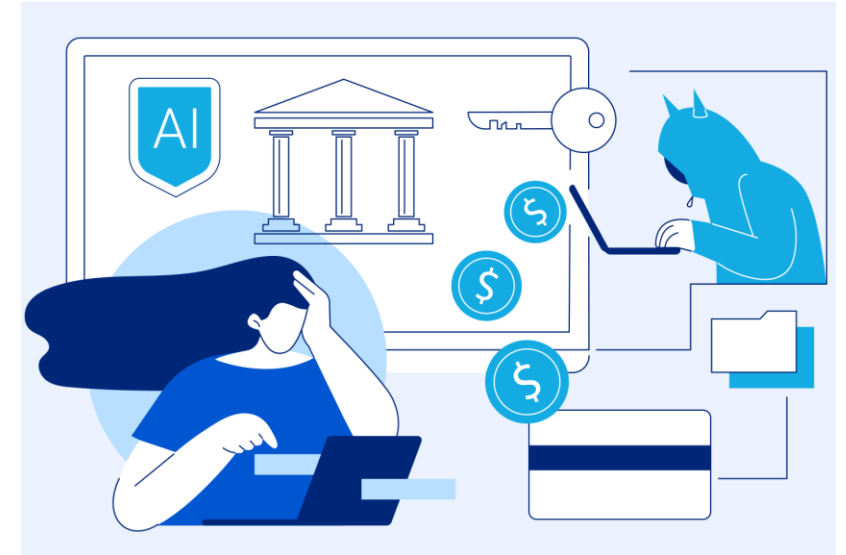
13. Algorithmic trading

- Investigate the applications of algorithmic trading, focusing on how algorithms optimize trading strategies, enhance market efficiency, and minimize risks.
- Explore the key challenges, including market volatility, regulatory compliance, and the potential for unintended consequences in algorithmic decision-making.
- Related work:
 - Chaboud, A. P., Chiquoine, B., Hjalmarsson, E., & Vega, C. (2014). Rise of the machines: Algorithmic trading in the foreign exchange market. *The Journal of Finance*, 69(5), 2045-2084.
<https://doi.org/10.1111/jofi.12186>
 - Treleven, P., Galas, M., & Lalchand, V. (2013). Algorithmic trading review. *Communications of the ACM*, 56(11), 76-85.
<https://doi.org/10.1145/2500117>



14. AI and fraud detection

- Explore the use of AI in fraud detection, focusing on its ability to identify fraudulent patterns, enhance security, and reduce financial losses.
- Address the challenges of implementing AI in fraud detection, including data quality, model accuracy, scalability, and regulatory concerns.
- Related work:
 - Bao, Y., Hilary, G., & Ke, B. (2022). Artificial intelligence and fraud detection. *Innovative Technology at the Interface of Finance and Operations*: Volume I, 223-247. https://doi.org/10.1007/978-3-030-75729-8_8
 - Abdallah, A., Maarof, M. A., & Zainal, A. (2016). Fraud detection system: A survey. *Journal of Network and Computer Applications*, 68, 90-113. <https://doi.org/10.1016/j.jnca.2016.04.007>



15. AI and credit scoring

- Explore the use of AI in credit scoring, focusing on its potential to improve accuracy, efficiency, and risk assessment in lending decisions.
- Investigate the challenges associated with AI-driven credit scoring, including data privacy, model fairness, transparency, and regulatory compliance.
- Related work:
 - Dastile, X., Celik, T., & Potsane, M. (2020). Statistical and machine learning models in credit scoring: A systematic literature survey. *Applied Soft Computing*, 91, 106263.
<https://doi.org/10.1016/j.asoc.2020.106263>
 - Markov, A., Seleznyova, Z., & Lapshin, V. (2022). Credit scoring methods: Latest trends and points to consider. *The Journal of Finance and Data Science*, 8, 180-201.
<https://doi.org/10.1016/j.jfds.2022.07.002>



➤ Seminar organization

- 2 SWS
- Maximum number of students: 15
- Each student select a topic
 - For each topic, 1-2 papers are provided as the starting point
 - You need to find additional papers on the topic by yourself
- Submission and grading
 - Oral presentation (35%)
 - Written report (65%)

Topic assignment

- Send an email to the instructor of the seminar module with the titles of two topics
 - 1st preference topic and a 2nd preference topic, until 18.04.2025
- In the email, please indicate which semester you are studying and attach your transcript as well.
- If there are more interested students than seminar places, the seminar places will be allocated by lottery, with students in the highest semesters having a higher chance of being assigned a topic.

- Presentation of the key points of the selected topic
 - Be prepared to answer relevant questions from the audience
- Timeline
 - 20 min presentation + 10 min discussion and 5 minutes feedback
- Participation in the discussion is very important from all seminar participants!

Written report

- Presentation of the topic, **critically** synthesising the information from the various literature sources you found
- The report is not expected to present new scientific findings, but to **review** and **reflect** on existing work
- 10 - 12 pages written in Latex formatted according to the Springer LNCS template*
- 50% of the references should be published in the last 3 years.

* Latex template: <https://www.overleaf.com/latex/templates/springer-lecture-notes-in-computer-science/kzwwwpvhwnvfj>

Timeline

It is important not to miss any of the below deadlines!

Tasks	Deadlines
Inquiries for seminar topic by email	Friday, 18.04.2025
Assignment of seminar topics	Friday, 25.04.2025
Submitting a table of content - Structure of report/presentation - References which you want to use	Friday, 09.05.2025
Submitting the first draft for peer review	Friday, 11.07.2025
Submitting the review for your peer draft	Friday, 18.07.2025
Feedback on slides (optional)	Friday, 01.08.2025
Seminar final presentation	TBA (mid of August)
Submitting the final written report	Friday, 29.08.2025

- Access code: seminar2025

 Seminar: AI in Finance

 Module overview

 Seminar materials

 Peer reviews

 Forum

 Seminar "AI in Finance: Threats and Challenges" SS 2025

Artificial Intelligence (AI) is revolutionizing finance, but it also introduces significant threats and challenges. The potential threats include, but are not limited to, fraud, algorithmic bias, and data security issues. By understanding these challenges, financial institutions, regulators, and tech developers can build strategies to mitigate risks, enhance transparency, and ensure fairness in financial systems. The seminar focuses on the potential threats and challenges of AI in the financial sector as well as exploring strategies to address these issues.

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 Module overview

Overview of the seminar and topics

<https://olat.vcrp.de/url/RepositoryEntry/4871159906>

- Presenting work or ideas from another source as your own, by incorporating it into your work (presentation and written report) without full acknowledgement, is considered plagiarism.
- Using graphs, charts, figures, or images from a source without acknowledging the latter is considered plagiarism as well.
- This will result in you failing the seminar!

All submission are going to be checked for plagiarism

- Scope of AI Use
 - While the seminar module permits the usage of AI tools to enhance learning and research abilities, understanding their appropriate usage is crucial.
 - These tools can assist in expediting data analysis, offering insights, and performing routine tasks, among other things.
- Acknowledgement of AI Use
 - When AI tools are employed in any portion of a student's research, report, or assignment, it is mandatory to explicitly acknowledge them.
 - This includes citing the specific AI tools used, their roles, and the extent of their contribution to the assignment.
 - Failure to do so could lead to academic penalties.

- Responsible AI Use
 - The use of AI tools to reproduce pre-existing results or automate time-consuming processes is acceptable. However, students should not rely on AI to generate original ideas, formulate academic arguments, or create independent coursework.
 - AI tools should supplement learning, not be a replacement for the cognitive process and individual creativity.
 - Keep in mind the limitations of AI
 - GPT-based chatbots are not always 100% accurate. Relying solely on these tools can risk misinformation or misunderstanding.
 - Dependency: There's potential risk that someone might become overly dependent on these tools, which could harm his/her development of individual research, writing skills, and critical thinking abilities. All assignments undergo thorough checks for both plagiarism and the use of AI assistance.

- Academic Integrity and Ethical Considerations
 - While AI tools can deliver advantages in terms of efficiency and accuracy, it is up to the user to use these tools ethically.
 - Plagiarism, inappropriate use, or any form of academic dishonesty will not be tolerated. Misrepresenting AI assistance as personal, original work is a serious offence that may lead to academic and disciplinary actions.
- Continuous Learning
 - The AI field is constantly evolving. Therefore, students are encouraged to stay up-to-date and explore new AI tools related to the seminar topic, by following research papers, attending seminars, webinars, or workshops to improve their understanding of the tool's capabilities and limitations.
- This AI policy strives to create a fair, ethical, and innovative learning environment for all participants in the seminar module. Adherence to these guidelines will ensure a rich and beneficial academic experience

Please note, any violation of this policy is subject to review by the course instructor

➤ Questions?